Functions:

Riboflavin is a component of two important coenzymes, namely flavin mono nucleotide (FMN) and flavin adenine dinucleotide (FAD). They play key roles in various enzymatic reactions.

FMN and FAD are synthesized as follows.

$$\text{Riboflavin} + \text{ATP} \rightarrow \text{FMN} + \text{ADP}$$

$$\text{FMN} + \text{ATP} \rightarrow \text{FAD} + \text{PP}_i$$

FMN and FAD combine with different apoenzymes to form a large number of redox enzymes.

Eg. FMN is associated with the enzyme cytochrome c reductase.

FAD is found in xanthine oxidase and acyl CoA dehydrogenase

Riboflavin is essential for a healthy skin and for good vision in bright light.

Absorption and storage

The vitamin is phosphorylated in the intestinal mucosa during absorption. It is absorbed by the small intestine through the portal vein and is distributed to all tissues. The major part is excreted in urine and a small part is metabolized in the body.

Deficiency:

Riboflavin deficiency leads to cheilosis. It is characterised by the development of fissures developing in the lips and at the corners of the mouth.



Figure 8.17 Deficiency due to vitamin B_2

Riboflavin is extremely sensitive to strong light. Due to sensitivity, riboflavin deficiency may occur in new born infants with hyper bilirubinemia who are treated by phototherapy.

iii. Vitamin B3 (Niacin)

Vitamin B_3 which is also called as niacin or nicotinic acid is pyridine 3-carboxylic acid. It occurs in tissues as nicotinamide (Figure 8.18).

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